

# segregation: Segregation Analysis, Inference, and Decomposition in Python

Renan X. Cortes<sup>1\*</sup>, Elijah Knaap<sup>2\*</sup>, and Sergio J. Rey<sup>3\*</sup>

<sup>1</sup> Federal University of Rio Grande do Sul <sup>2</sup> University of California, Irvine <sup>3</sup> San Diego State University  
\* These authors contributed equally.

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

## Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Henrik Finsberg](#) 

## Reviewers:

- [@mandarc64](#)
- [@Sidrcs](#)

Submitted: 29 June 2026

Published: unpublished

## License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

## Summary

segregation is an open-source Python library for measuring, analyzing, inferring, and decomposing social and spatial segregation. Developed within the PySAL ecosystem, it provides a comprehensive framework for quantifying segregation across demographic, socioeconomic, and geographic contexts using both aspatial and spatially explicit measures. The package implements over 40 single-group, multigroup, local, and spatial segregation indices, capturing dimensions such as evenness, exposure, concentration, clustering, centralization, and spatial proximity. It supports multiple spatial representations, including adjacency, distance-based, and network-based relationships.

Beyond point estimation, segregation provides Monte Carlo inference for single-value and comparative hypothesis testing under alternative null models, as well as decomposition methods that separate differences in segregation into demographic composition and spatial structure. Built for Pandas and GeoPandas, segregation integrates seamlessly with the PySAL ecosystem ([S. J. Rey et al., 2022](#); [S. J. Rey & Anselin, 2007](#)), providing transparent, reproducible, and extensible tools for segregation research and spatial data science.

## Statement of need

Residential segregation is a central topic in urban studies, sociology, demography, geography, and public policy. Although demographic and spatial data are increasingly available, existing software often provides a limited set of segregation measures, relies on proprietary platforms, or lacks support for modern inference and decomposition methods. Within Python, segregation analysis has historically depended on custom implementations despite the widespread adoption of Pandas and GeoPandas.

segregation addresses these limitations by providing:

- A comprehensive, well-documented API for segregation analysis
- Native integration with Pandas, GeoPandas, and the PySAL ecosystem
- Over 40 aspatial, spatial, local, and multigroup segregation indices
- Multiscalar segregation profiles
- Simulation-based inference for hypothesis testing
- Decomposition of segregation into demographic and spatial components
- Transparent, reproducible, and extensible implementations

These capabilities establish segregation as a unified, open-source framework for reproducible segregation analysis in Python.

## State of the field

segregation is part of the PySAL ecosystem, where it provides tools for residential segregation analysis within the **Explore** family of spatial analysis libraries. Unlike desktop GIS software, it supports reproducible, scriptable workflows that integrate naturally with broader Python data science pipelines.

Although segregation analysis is also available in R packages such as *OasisR* (Tivadar, 2019), *seg* (Hong, 2011), and *segregation* (Elbers, 2021), PySAL's segregation module provides a broader and more integrated framework. It includes over 40 segregation indices, supports multiscalar and network-based analyses, Monte Carlo inference, and decomposition methods, while enabling simultaneous computation of multiple indices with minimal software dependencies. As a native Python library, it integrates seamlessly with the PySAL ecosystem and modern geospatial data science workflows.

## Software design

segregation is designed with attention to flexibility, usability, and interoperability between its core functions. The library is organized into different subparts such as *singlegroup*, *multigroup*, *local*, *inference*, *decomposition*, *batch*, *network*, and *dynamics*.<sup>1</sup> Segregation measures are built using Python classes which can be integrated in subsequent steps, such as inference or decomposition. The module is structured toward two kinds of segregation indices: 'spatially-explicit' and 'spatially-implicit'. The former includes space as part of its original formula. The latter uses the Sean F. Reardon & O'Sullivan (2004) approach to state that any segregation index is a spatial index if you transform the data properly.

Additionally, segregation is developed with testing and documentation standards consistent with the Scientific Python ecosystem, ensuring reliability and maintainability.

## Core Functionality

segregation organizes its functionality around the type of segregation analysis the user is interested in, and each subpart is explained as follows.

### Single and Multigroup Indices

Single-group measures assess segregation between two different groups in a given location (i.e., one group vs. everyone else). Multigroup segregation evaluates the simultaneous separation of all groups in a population (e.g., the distribution of White, Black, Asian, and Hispanic residents) across areas.

Currently, segregation has over 40 indices available which represents, to our knowledge, the broadest range of indices available for a user in any software. Also, the user can fit many indices at once with a wrapper function in the batch module.

### Local Indices

Unlike global indices that summarize an entire metropolitan area into a single value, local indices decompose segregation to the individual geographic unit level. Using these disaggregated measures helps identify precise spatial clusters where social isolation is most acute, uncovering micro-level dynamics that global metrics often mask. Currently, segregation has seven local indexes.

<sup>1</sup>Cortes et al. (2020) introduced segregation but many new implementations were developed recently and the API underwent a major revision.

## 78 Multiscalar

79 The multiscalar profile (Sean F. Reardon et al., 2008) is a tool for measuring spatial segregation  
80 dynamics—the way that a segregation index changes values as the concept of a neighborhood  
81 changes, and what that tells us about macro versus micro patterns of segregation. The core  
82 idea is to calculate a segregation statistic, then expand the spatial scope of a neighborhood,  
83 recalculate the statistic, and repeat.

84 The package has a wrapper named `compute_multiscalar_profile` which can be used in a  
85 workflow to build these profiles.

## 86 Simulation based Inference

87 PySAL's segregation module provides Monte Carlo inference for evaluating the statistical  
88 significance of segregation indices under different null hypotheses. For single-value  
89 inference, it supports resampling approaches such as bootstrap, systematic, evenness, and  
90 geographic\_permutation. For comparative inference, it includes methods such as bootstrap  
91 and composition, which generate synthetic distributions through counterfactual estimates.  
92 Because different null hypotheses test distinct assumptions, their specification is critical  
93 and can lead to substantially different conclusions. Likewise, not all segregation indices are  
94 appropriate for every null hypothesis, particularly in comparative analyses, making careful  
95 selection of both the index and inference procedure essential.

## 96 Decomposition

97 The PySAL segregation module implements a decomposition framework for comparative  
98 segregation analysis that partitions differences in segregation into population composition and  
99 spatial structure. Building on S. Rey et al. (2021), it combines spatially explicit counterfactual  
100 distributions with Shapley value decomposition to quantify each component's contribution to  
101 differences in segregation across cities, time periods, and spatiotemporal contexts. Applicable  
102 to multiple segregation indices, the framework provides a richer interpretation than direct  
103 index comparisons by identifying whether observed differences primarily reflect demographic  
104 composition or neighborhood spatial organization. This functionality is available through the  
105 `DecomposeSegregation` class in the decomposition subpart.

## 106 Example workflow

107 Assuming a pandas or geopandas DataFrame named `data_1`, a segregation index can be  
108 computed as:

```
109 from segregation.singlegroup import Dissim  
  
110 seg_index = Dissim(  
    data_1,  
    group_pop_var="group_A",  
    total_pop_var="total_population",  
)
```

109 The estimated value is available through the `statistic` attribute. Statistical significance can  
110 be evaluated using Monte Carlo inference:

```
111 from segregation.inference import SingleValueTest  
  
112 result = SingleValueTest(seg_index, null_approach="bootstrap")
```

111 The package also provides wrappers for computing all single-group indices (`batch_compute_singlegroup`),  
112 generating multiscalar profiles (`compute_multiscalar_profile`), comparing segregation  
113 measures (`TwoValueTest`), and decomposing differences into demographic and spatial  
114 components (`DecomposeSegregation`).

## Research impact statement

segregation is actively used in research on urban inequality, spatial demography, and education policy. Applications include comparative analyses of racial segregation (Cortes et al., 2020), measurement of spatial information theory indices across U.S. metropolitan areas (Knaap & Rey, 2024), studies of school-neighborhood segregation (S. J. Rey et al., 2024), optimization approaches for reducing school segregation (Wei et al., 2022), comparative segregation analytics (S. Rey et al., 2021), and analyses of historical redlining legacies (S. Rey & Knaap, 2022).

The package is also used in spatial data science education, including textbooks (Knaap, 2026) and instructional materials presented at conferences such as SciPy.<sup>2</sup>

## AI usage disclosure

No generative AI or LLMs were used for code development in segregation; however, they were used for grammar, spelling corrections and consistency during the writing of this paper.

## Acknowledgements

segregation is developed as part of the PySAL community, which brings together researchers and developers working on spatial analysis methods and software. The project builds on decades of research in segregation, urban, and spatial data science, and benefits from contributions across the open-source geospatial community.

Funding from National Science Foundation Grants 2345820 and 1831615 have supported segregation development.

The following acknowledgement applies to Renan X. Cortes:

Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) through process 88881.170553/2018-01 have supported segregation development.

## References

- Cortes, R. X., Rey, S., Knaap, E., & Wolf, L. J. (2020). An open-source framework for non-spatial and spatial segregation measures: The PySAL segregation module. *Journal of Computational Social Science*, 3(1), 135–166. <https://doi.org/10.1007/s42001-019-00059-3>
- Elbers, B. (2021). A method for studying differences in segregation across time and space. *Sociological Methods & Research*, 52(1), 5–42. <https://doi.org/10.1177/0049124121986204>
- Hong, S.-Y. (2011). *Seg: Measuring spatial segregation*. <https://cran.r-project.org/web/packages/seg>
- Knaap, E. (2026). *Urban analysis & spatial science*. [https://knaaptime.com/urban\\_analysis](https://knaaptime.com/urban_analysis)
- Knaap, E., & Rey, S. J. (2024). Segregated by design? Street network topological structure and the measurement of urban segregation. *Environment and Planning B: Urban Analytics and City Science*, 51(7), 1408–1429. <https://doi.org/10.1177/23998083231197956>
- Reardon, Sean F., Matthews, S. A., O'Sullivan, D., Lee, B. A., Firebaugh, G., Farrell, C. R., & Bischoff, K. (2008). The geographic scale of metropolitan racial segregation. *Demography*, 45(3), 489–514. <https://doi.org/10.1353/dem.0.0019>

<sup>2</sup><https://www.youtube.com/watch?v=4AHJVMs7iH4>

- 154 Reardon, Sean F., & O'Sullivan, D. (2004). Measures of spatial segregation. *Sociological*  
155 *Methodology*, 34(1), 121–162. <https://doi.org/10.1111/j.0081-1750.2004.00150.x>
- 156 Rey, S. J., & Anselin, L. (2007). PySAL: A Python Library of Spatial Analytical Methods. *The*  
157 *Review of Regional Studies*, 37(1), 5–27. <https://doi.org/10.52324/001c.8285>
- 158 Rey, S. J., Anselin, L., Amaral, P., Arribas-Bel, D., Cortes, R. X., Gaboardi, J. D., Kang, W.,  
159 Knaap, E., Li, Z., Lumnitz, S., Oshan, T. M., Shao, H., & Wolf, L. J. (2022). The **PySAL**  
160 Ecosystem: Philosophy and Implementation. *Geographical Analysis*, 54(3), 467–487.  
161 <https://doi.org/10.1111/gean.12276>
- 162 Rey, S. J., Knaap, E., Wei, R., & Skrah, D. (2024). Measuring Spatial Congruence in the School-  
163 Neighborhood Nexus. *Applied Spatial Analysis and Policy*. [https://doi.org/10.1007/s12061-](https://doi.org/10.1007/s12061-024-09603-9)  
164 [024-09603-9](https://doi.org/10.1007/s12061-024-09603-9)
- 165 Rey, S., Cortes, R., & Knaap, E. (2021). Comparative spatial segregation analytics. *Spatial*  
166 *Demography*, 9, 1–26. <https://doi.org/10.1007/s40980-021-00075-w>
- 167 Rey, S., & Knaap, E. (2022). The Legacy of Redlining: A Spatial Dynamics Perspective.  
168 *International Regional Science Review*, 0(0). <https://doi.org/10.1177/01600176221116566>
- 169 Tivadar, M. (2019). OasisR: An R package to bring some order to the world of segregation  
170 measurement. *Journal of Statistical Software*, 89(1), 1–39. [https://doi.org/10.18637/jss.](https://doi.org/10.18637/jss.v089.i07)  
171 [v089.i07](https://doi.org/10.18637/jss.v089.i07)
- 172 Wei, R., Feng, X., Rey, S., & Knaap, E. (2022). Reducing Racial Segregation of Public School  
173 Districts. *Socio-Economic Planning Sciences*, 101415. [https://doi.org/10.1016/j.seps.](https://doi.org/10.1016/j.seps.2022.101415)  
174 [2022.101415](https://doi.org/10.1016/j.seps.2022.101415)